Vraj Patel

Software Engineering, Data Engineering and Applied ML

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Build software and research data systems that connect backend engineering, dependable pipelines, and applied ML. Work spans CU Boulder internal tools, parliamentary archives, speech processing, and team research projects.

EXPERIENCE

Systems Integration Engineer
May 2025 - Present

CU Boulder | Institute of Behavioral Science

- Build a FastAPI, React/TypeScript, and PostgreSQL support-operations platform around 10,803 reconciled Freshdesk tickets in the August 2026 snapshot.
- Designed incremental source reconciliation with durable watermarks, idempotent upserts, lifecycle-aware records, and fail-closed readiness checks.
- Trained an unreleased BERT routing candidate on 6,137 tickets: 82.63% sealed-test accuracy and 0.7563 macro-F1, above the prior XGBoost baseline by 3.11 and 5.36 points.
- Reduced the analytics visualization payload from 361 KB to 123 KB gzip (66%) by replacing Plotly with Recharts and accessible chart tables.
- Automated Active Directory offboarding with PowerShell, reducing typical group-cleanup effort from about 15 minutes to under one minute per user.

Research Assistant, Data Engineering
Jan 2025 - May 2026

Princeton University | Indian parliamentary research

- Built fault-tolerant Selenium collection workflows for Indian parliamentary archives spanning 1951-2024, with pagination and resumable SQLite state.
- Extracted thousands of parliamentary PDFs with PyMuPDF and standardized inconsistent ministry names with fuzzy matching and reviewable intermediate data.

Lead Technical Research Assistant
Mar 2023 - May 2025

CU Boulder | Institute of Behavioral Science

- Built a Python/AWS speech pipeline using Amazon Transcribe, Translate, and S3; the combined campaign collection covered 200+ hours, with 181 speeches in the 2024 run.
- Adapted Amazon Transcribe with a custom language model trained on about 110K Hindi words; WER fell from 40.1% to 30.1% on one held-out speech, not a corpus-wide benchmark.
- Built PyTorch/ResNet50 embedding and k-means analysis workflows with researchers studying visual bias in COVID-19 news imagery.

Undergraduate Research Assistant, Satellite Telemetry
Aug 2024 - May 2025

The Data Mine of the Rockies | Purdue and L3Harris

- Trained Isolation Forest and LSTM models on synthetic NOS3 satellite telemetry to detect and classify simulated cyberattack anomalies.
- Presented satellite cybersecurity research with a nine-student, four-university team; mapped scenarios to SPARTA and MITRE ATT&CK.

Research Assistant, Data Engineering and ML
Jan - Apr 2024

CU Boulder | Studio Lab

- Compared six Hindi speech-to-text engines with word-error-rate evaluation and assessed translation services; recommended Amazon Transcribe for the later IBS pipeline.

PROJECTS

SeeMyRace | Race Photo Search
Flask, PostgreSQL/pgvector, Docker

- Built authenticated, user-scoped match verification and GPX-backed race creation for a team race-photo application; authored 32 commits across 24 paths through seven merged PRs.
- Implemented upload/search and admin workflows, and evaluated OCR/SWT approaches; later face/OCR model and retrieval changes include work by teammates.

NetPrice Finder | Price Comparison
FastAPI, Redis, Playwright, Docker Compose

- Built a multi-service price-comparison system with a REST API, browser scraper, Redis caching, database persistence, container health checks, and an internal service network.

HateBlocker | Classical NLP Models
scikit-learn, XGBoost, TF-IDF

- Owned classical-ML exploration, 5,000-term TF-IDF features, 17 stylometric features, and grid-searched baselines in a five-person project; XGBoost reached 90.72% held-out accuracy.
- Evaluated three classes on 2,479 held-out examples; hateful-class recall remained 0.24. Teammates owned BERT, the FastAPI service, and browser extension.

Automated Interview Scoring
SentenceTransformers, scikit-learn, TensorFlow

- Team project using MiniLM alignment, lexical and 35 prosodic features, mutual-information selection, k-fold regressors, and SHAP analysis for 138 mock interviews.

Diabetes Risk Modeling
TensorFlow, Keras Tuner, scikit-learn, SHAP

- Built SVM and neural-network models plus SHAP analysis in a three-person project using the balanced 70,692-record BRFSS diabetes sample; compared results with teammates' baselines.

REST and gRPC Benchmark
Python, Protobuf, GCP

- Built and benchmarked REST and gRPC services on GCP VMs, comparing cross-region raw-image calls and persistent-connection reuse.

vrajmpatel.com | Portfolio and Delivery Pipeline
Astro, TypeScript, GitHub Actions, Cloudflare Workers

- Built static-site CI/CD with automated tests, dependency audit/review gates, SHA-pinned actions, a public-asset privacy scan, and verified GitHub Pages artifacts.

Hybrid Home and Cloud Infrastructure
Terraform, Oracle Cloud, Docker Compose, Tailscale

- Operate a Terraform-defined cloud node and local container services with private remote access, internal database networking, and reproducible deployment configuration.

EDUCATION

M.S. Computer Science + M.E. Engineering Management — Expected May 2028
University of Colorado Boulder | Concurrent dual degree

B.S. Computer Science, Business Minor — May 2025
University of Colorado Boulder | GPA: 3.84 technical, 3.65 overall

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++, Bash, PowerShell
Software: FastAPI, Flask, React, SQLAlchemy, REST, gRPC, pytest
Data: PostgreSQL, SQLite, Redis, pgvector, Pandas, source reconciliation, data validation
ML: PyTorch, scikit-learn, Transformers, TensorFlow, NLP, model evaluation, SHAP
Infrastructure: Docker, Kubernetes/OpenShift, AWS, GCP, Terraform, Linux, GitHub Actions